

Question\_7:

Code:

```
1  /*
2  Question_7:
3  Find the minimum sum of A[i] * B[i] after shuffling.
4  */
5
6  #include <iostream>
7  #include <algorithm>
8  using namespace std;
9
10 void sort(int a[], int n, int flag) {
11     for (int i = 0; i < n - 1; i++)
12         for (int j = 0; j < n - i - 1; j++)
13             if ((flag && a[j] > a[j + 1]) ||
14                 (!flag && a[j] < a[j + 1]))
15                 swap(a[j], a[j + 1]);
16 }
17 int main() {
18     int T;
19     cin >> T;
20
21     while (T--) {
22         int n;
23         cin >> n;
24
25         int a[n], b[n];
26         for (int i = 0; i < n; i++) cin >> a[i];
27         for (int i = 0; i < n; i++) cin >> b[i];
28
29         sort(a, n, 1);
30         sort(b, n, 0);
31     }
```

```
32         long long ans = 0;
33         for (int i = 0; i < n; i++)
34             ans += 1LL * a[i] * b[i];
35
36         cout << ans << endl;
37     }
38
39     return 0;
40 }
```

Output:

Test Case\_1:

```
3
4
2 5 3 7
8 5 7 3
83
3
13 11 1
6 15 14
247
5
6 11 9 15 4
3 41 8 21 4
451
PS C:\Users\Prabin Yadav\
```

**Test Case\_2:**

```
2
3
13 11 1
6 15 14
247
5
6 11 9 15 4
3 41 8 21 4
451
PS C:\Users\Prabin Yadav
```